

Air-to-Ground Transmission of Time-Encoded Qubits with Single-Mode Fiber Coupling for Terrestrial Network Compatibility

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Abstract We present a demonstration of time-encoded quantum bit transmission over an air-to-ground link between a mobile airborne transmitter and an optical ground station. Our results pave the way towards constructing secure and flexible hybrid Quantum Key Distribution links beyond currently available fiber-optic topologies. ©2026 The Author(s)

Introduction

Quantum Key Distribution (QKD) is nowadays a mature and well-developed measure against potential adversaries equipped with quantum computers [1]. It enables information-theoretic secure generation of cryptographic keys compatible with encryption standards resistant against modern cyberthreats [2]. Fiber-based QKD has received particular attention because of the broad availability of terrestrial fiber infrastructure, the maturity of optical networking technology, and the relatively low propagation losses of fiber links [3, 4]. Consequently, numerous QKD demonstrations and network topologies, including hybrid architectures, have been realized [5].

However, a typical feasible distance of a single point-to-point connection over a deployed fiber typically lies between around 100-400 km, depending on the detectors and infrastructure [3]. This enforces using multiple trusted nodes for implementing desired topologies and extending the link [6], effectively increasing the cost hardware requirements. Moreover, the security of the entire connection depends on the integrity of every trusted node. Therefore, compromising any one of them may result in complete secret leakage. That weakness can be overcome by engineering alternative trusted routes [7]. For this reason, free-space and satellite quantum links are being actively investigated as complementary solutions compatible with fiber-based networks. Free-space QKD has already been demonstrated in terrestrial [8], aerial, and satellite scenarios [9], with recent progress including a quantum link established via an unmanned aerial vehicle (UAV) [10]. The field of space and aerial quantum communication witnessed little demonstrations so far. QKD with an aircraft has been demonstrated using BB84 protocol based on polarization encoding combined with multi-mode-coupled

receivers at a distance of approx. 20 km with a QBER of 4.8 % [11]. This multi-mode receiver approach leaves open the door for spatial mode mismatch attacks [12]. In contrast, single-mode fiber coupling removes this vulnerability and is compatible with terrestrial fiber networks. For example, this allows the flexible spatial separation of optical ground station (OGS) and QKD receiver

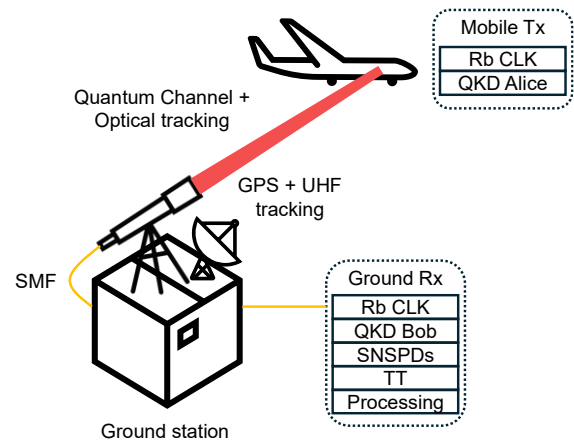


Fig. 1: Experimental setup. The rack with transmitter (Tx) equipment was placed in a flying aircraft and included the QKD transmitter (QKD Alice) and the rubidium clock (Rb CLK) for the reference. The repeated pattern of time-encoded qubits was transmitted over a free-space channel to the receiving ground station, as it was tracking the airplane using the Global Positioning System (GPS), ultra-high frequency radio (UHF), and laser beacon beams. Single-photon level signals were coupled to the single-mode fiber (SMF) and directed to the receiver (Rx, QKD Bob). The signals were decoded using single photon counting, involving another rubidium clock, superconducting nanowire single-photon detectors (SNSPDs) and a time tagger (TT). Registered events were analyzed for the quantum bit error rate (QBER) and corrected for the clock drift with a processing server in real time.

such that the OGS can be placed for optimum visibility, while the QKD receiver can be placed inside a distant secure building.

In this paper we present a realisation of an air-to-ground link established from an airplane with our 1-decoy time-phase BB84 QKD system [13] featuring clock-channel free operation. We report on real-time quantum bit error rate (QBER) values measured during flight using our efficient qubit-based synchronisation method [14]. Our results demonstrate capability of fast secure link recovery in a dynamic scenario, paving the way towards long-distance mobile end-to-end key generation. To the best of our knowledge, this is the first demonstration of time-encoded qubit transmission from a moving platform over an aerial link operated without a dedicated clock channel combined with single-mode receiver coupling.

Experimental setup

The experimental setup is based on a time-phase discrete-variable QKD system comprising transmitter (Alice) and the receiver (Bob). The main parts of the setup are shown in Fig. 1.

We used the 1-decoy state BB84 time-phase protocol [15] operated at a repetition rate of 625 MHz. Temporal encoding was based on emitting strongly-attenuated optical pulses at a fixed wavelength forming weak coherent states in the middle of either early or late time bin, which was 800 ps wide. Only the temporal encoding was used during the experiment. The central wavelength was set at 1546.92 nm (ITU C38 DWDM channel). The transmitter emitted a fixed pattern of qubits at two different intensities in the time domain, effectively encoding signal and decoy states at mean photon numbers of $\mu_1 = 0.47$ and $\mu_2 = 0.17$, respectively. The qubits were transmitted over a free-space channel to the ground

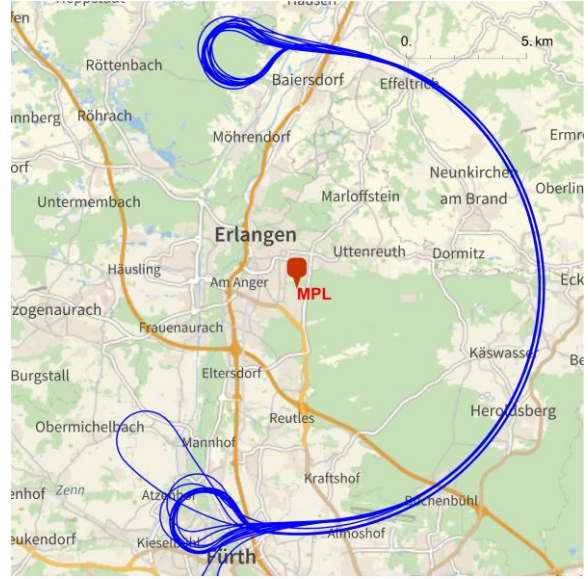


Fig. 2: Airplane trajectory during the flight campaign. The aircraft flew in semi-circles at a fixed altitude of 3 km around the ground station at the Max Planck Institute for the science of light (MPL). The effective link distance was approx. 10 km.

station, and further to the laboratory over a single-mode fiber. Transmitter and receiver were disciplined by two independent, free-running Rubidium clocks (Safran RBSources 1500), each generating a 10 MHz sinewave reference signal.

To suppress detrimental stray light from the sun and the beacon lasers on the receiver side, we used three cascaded 100 GHz band-pass filters centred around ITU C38 DWDM channel with a combined suppression of approx. 120 dB. We registered the qubits with superconducting nanowire single-photon detectors (SNSPDs, SingleQuantum, $60\% \pm 20\%$ efficiency, dark count rate < 50 Hz) and a time tagger (Swabian Instruments Time Tagger Ultra with a resolution below 10 ps). The detection events were post-

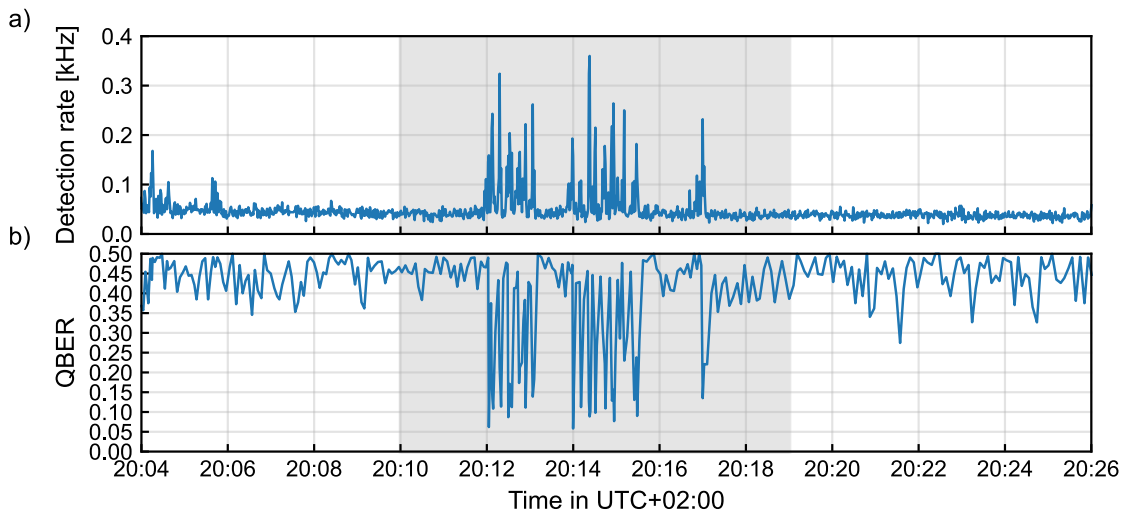


Fig. 3: Experimental results: a) Single-photon signal detection rate b) Measured QBER. The gray area marks the interval where statistics indicate a successful link acquisition and qubit detection.

processed on a dedicated server with an aid of our syntonization algorithm in real time. The algorithm parameters were selected to compensate for both drifts between the free-running clocks, as well as drifts introduced by the Doppler effect. This drift correction was performed by applying a piecewise linear correction to blocks of 500 counts. After the drift correction, we used temporal filtering with a timebin-centered pass window of 25% and finally calculated the QBER.

Air-to-ground link

The link was established between an aircraft (Dornier DCFFU, operated and provided by the Deutsches Zentrum für Luft- und Raumfahrt (DLR)) flying in semicircles and the receiving OGS (QuBUS, operated and provided by the Fraunhofer Institute for Applied Optics and Precision Engineering (IOF)) relaying optical signals to the laboratory located at the Max Planck Institute for the physics of light in Erlangen, as shown in Fig. 2. The flight duration of one semi-circle was approximately 8 minutes. The aircraft was equipped with Global Positioning System (GPS) and an optical laser terminal capable of tracking a beacon laser for link acquisition and stabilization [11]. The transmitted laser beam from that system was multiplexed with our quantum signals from the on-board devices and transmitted via the optical antenna. The ground station was tracking the aircraft its GPS coordinates transmitted via ultra-high frequency radio (UHF) and an optical downlink beacon [8]. After coarse link acquisition, fine tracking algorithms and adaptive optics were used to establish and maintain single-mode fiber coupling. Albeit very challenging and less efficient in comparison to the multi-mode fiber coupling, single-mode confinement grants compatibility with terrestrial fiber networks and helps to suppress stray light propagating in the other modes. Furthermore, it allows placing the QKD receiver many kilometres away from the OGS. After fiber coupling in the OGS, quantum signals were demultiplexed and directed to the QKD receiver. Attenuation of the free-space channel was measured with a beacon laser operating in the C band. Another C-band tunable laser source (TLS) was used to measure the attenuation between the ground station and the receiver in the laboratory. The average attenuation was 48.0 ± 3.0 dB, as measured during the stable link time. Strong fluctuations were observed due to atmospheric turbulences and changing weather conditions. The impact of the beacon beams and residual stray light was measured with the SNSPDs as a detected countrate in various configurations. After sunset (20:00), the total background contribution remained below 15 Hz. During daytime

operation, however, the contribution consistently exceeded the detector dark count rate of 40 Hz under all weather conditions. Cloud scattering produced relatively high and spatially uniform stray-light levels. Under clear-sky conditions, the solar background showed a strong dependence on the field of view and pointing direction of the OGS telescope. However, even when the telescope was pointed away from the sun, reflections from the aircraft contributed significantly to the measured photon flux. In Fig. 3 we present registered countrates and corresponding QBER values. Strong fluctuations in values stem from turbulences and rapid action of tracking methods. Nevertheless, our system could typically recover synchronization within less than 1 s and therefore was able to perform real time measurements. Additional fluctuations may also arise from polarization drift, which strongly influences the detection efficiency of the SNSPDs. The QBER fell below 10.0 % multiple times, reaching the minimal value of 5.6 %. Although a full QKD experiment could not be performed because the classical channel was unavailable for technical reasons, the measured QBER values remain consistent with the thresholds required for secure QKD [16].

Conclusion and discussion

We demonstrated successful air-to-ground transmission of time-encoded qubits using our in-house QKD system and syntonization algorithm. We successfully measured QBER values below 10.0 %, with lowest value of 5.6 %, while coupling to a single mode fiber in an OGS from a moving airborne transmitter. Assuming the asymptotic scenario, a positive secure key rate could be achieved with these error rates [16]. Our system demonstrated robustness, flexibility and ease of operation, as data were processed in real time without the dedicated clock channel being needed for syntonisation. The algorithm proved effective in compensating any instabilities caused by the natural clock drift and the Doppler effect introduced by the moving transmitter. Our findings are an important step towards satellite-/free-space-based QKD links compatible with currently existing fiber infrastructure, paving the way to constructing secure hybrid links.

Acknowledgements

We gratefully acknowledge our partners at the DLR for organising the flight campaign, leading the flight certification process and operating the provided aircraft that served as the mobile transmitter platform for these experiments. We further thank our partners at the IOF for constructing, testing, and operating the optical ground station QuBUS and all the tracking mechanisms.

Combined expertise of the DLR and the IOF enabled establishing the free-space optical link, via which we could transmit the qubits. We also thank the Max Planck Institute for the Science of Light in Erlangen for hosting the flight campaign and granting us laboratory space as well as access to the SNSPDs. This research was conducted within the scope of the project QuNET, funded by the German Federal Ministry of Research, Technology and Space (BMFTR) in the context of the federal government's research framework in IT-security "Digital. Secure. Sovereign."

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